PENK Dry-Lab Modelling — Consolidated Project Brief

iGEM Patras · Dry-Lab / Modelling Prepared as a consolidated audit and design record across four computational workstreams. All methods, force fields, databases, and design rules are traceable to the accompanying methods & references log.

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0. Executive summary

This project supports an enkephalin-based analgesic strategy built on **proenkephalin (PENK)** expression plus a **neprilysin (NEP/MME) knockdown** arm, delivered by cationic/ionizable **lipoplexes**. The dry-lab work spans four independent but linked deliverables:

1. **Plasmid audit** of four PENK expression cassettes — identity, gene order, linker, and PENK ORF integrity verified against the native reference (UniProt P01210). **One real defect found** (plasmid 1, truncated PENK).
2. **Restriction-digest QC** — a two-tier digest scheme (insert release + identity fingerprint) with a virtual gel, designed to confirm construct identity before transfection.
3. **Human anti-NEP (MME) siRNA design** — four validated-in-silico siRNA duplexes targeting the CDS shared by all MME transcript variants, with off-target and thermodynamic scoring.
4. **Lipoplex CG-MD** — coarse-grained MARTINI simulations comparing three lipid formulations (permanent-cation DOTAP, protonated ionizable MC3-H^+, neutral MC3^0) across triplicate replicates, resolving the pH-dependent DNA-binding behaviour that motivates the ionizable-lipid choice.

Scope note on the payload. The four cassette plasmids carry the **native** human PENK sequence — they are the expression-system test-bed. The engineered anionic-address PENK (the therapy payload) is a separate construct: it currently exists as a **cloning-ready insert** (`engineered_PENK_GC-optimized.fasta` / `engPENK_insert_annotated.gb`), not yet assembled into an expression plasmid. It is audited here as the payload (§1) and kept out of the four-plasmid restriction QC (§2); §2b then assembles it into a therapy plasmid on the test-bed backbone with its own purpose-built QC.

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1. Plasmid audit — four PENK cassettes

All four plasmids are pcDNA3.1(+)-type backbones carrying native human PENK and mEGFP, differing in **gene order** (which gene is cap-driven / first) and **linker** (EMCV IRES vs self-cleaving P2A).

#	Plasmid	Size (bp)	Architecture	PENK identity	Cores	Verdict
1	mEGFP-IRES-PENK	7,500	CMV → mEGFP (cap) → IRES → PENK	**TRUNCATED (263 aa)**	6 of 7	**DEFECT**
2	mEGFP-P2A-PENK	6,971	CMV → mEGFP–P2A–PENK (single ORF)	full length (266 aa)	7	OK, caveat
3	PENK-IRES-mEGFP	7,512	CMV → PENK (cap) → IRES → mEGFP	full length (267 aa)	7	**OK — cleanest**
4	PENK-P2A-mEGFP	6,971			7	OK

#	Plasmid	Size (bp)	Architecture	PENK identity	Cores	Verdict
			CMV → PENK–P2A–mEGFP (single ORF)	full length (267 aa)		
—	**engineered PENK (payload)**	885 (insert)	NheI–Kozak–ATG–CDS–XhoI (cloning-ready insert)	**engineered** (288 aa)	**7 of 7**	OK — payload

Key findings

- **Plasmid 3 (PENK-IRES-mEGFP) is the cleanest construct**: PENK is cap-dependent with an intact signal peptide (MARFLTLC…), byte-identical to the UniProt P01210 reference.
- **Plasmid 1 has a genuine defect**: a premature stop truncates the C-terminal **YGGFMRF** (the 7th enkephalin core), leaving 6 of 7 cores. This is a point-level lesion — **it will not change any restriction-gel band** and can only be caught by Sanger sequencing of the PENK 3′ end.

- **Plasmid 2 caveat**: PENK is downstream of P2A, so its N-terminus becomes P-ARFLTLC (loses initiator Met, gains a P2A-derived proline) — this may impair signal-peptide function and secretion. For a secreted peptide product, the secreted gene should be placed **first** (as in plasmids 3 and 4).

- **Engineered payload (therapy construct)**: the anionic-address PENK, in which all seven enkephalin cores are recoded YGGF[M/L] → **YGGFEVVG**. Verified from `engineered_PENK_GC-optimized.fasta`: 867 nt CDS → 288 aa, Met start, single clean stop, **7 of 7 engineered cores, 0 native cores**, GC 59.9%, no homopolymers, cores made synonymously distinct, and single flanking NheI/XhoI sites with **no internal cut sites** (cloning-clean for pcDNA3.1(+)). This is the construct that gives the therapy its function; it is **not yet in a backbone**, so it appears in the audit but not in the four-plasmid gel QC.

Full per-feature detail is in `plasmid_feature_audit.csv`.

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2. Restriction-digest QC protocol

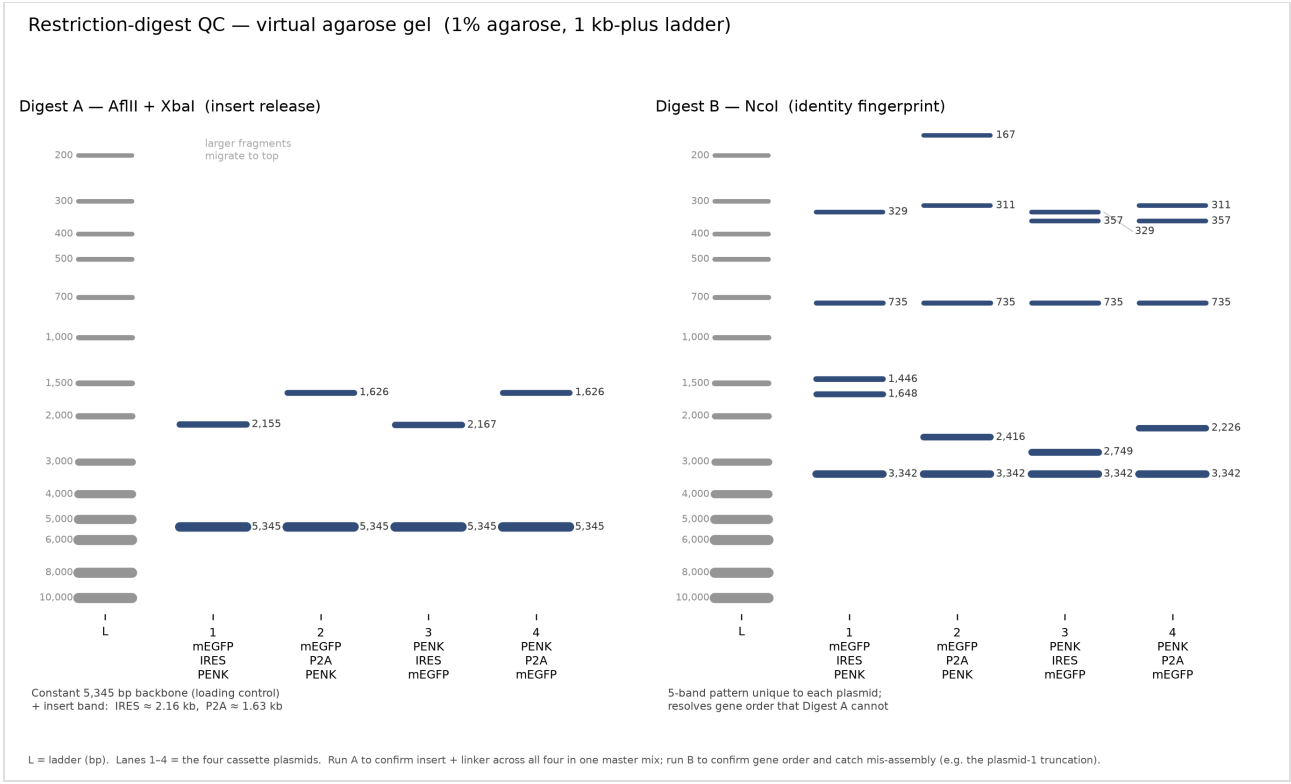
A two-tier scheme confirms both **linker type** and **gene order**, and is designed so a single defect (mis-assembly) is caught before transfection.

Digest A — insert release (AflII + XbaI). Releases a constant **5,345 bp backbone** band (loading / complete-digestion control) plus one insert band whose size reports the linker: **~2.16 kb = IRES**, **~1.63 kb = P2A**. Limitation: cannot distinguish the two IRES plasmids (12 bp apart) or the two P2A plasmids — gene order requires Digest B.

Digest B — identity fingerprint (NcoI, 5 cuts). A unique 5-band pattern per plasmid. Shared 3,342 + 735 bp landmarks; the **second-largest band is diagnostic** for gene order (1,648/1,446 doublet vs 2,416 vs 2,749 vs 2,226 bp).

Critical limitation flagged in the protocol: passing the gel QC does **not** clear plasmid 1 — the PENK truncation is invisible on a gel and must be confirmed by **Sanger sequencing**.

The virtual gel (`restriction_qc_gel.png`) shows both digests side-by-side against a 1 kb-plus ladder; the full bench recipe and interpretation logic are in `restriction_qc_protocol.md`.



Virtual restriction gel — four native PENK plasmids, Digest A (AflII+XbaI) and Digest B (NcoI) vs 1 kb-plus ladder.

Applies to the four native plasmids only. This scheme is native-PENK-specific: the engineered payload insert has **no AflII and no XbaI site** (it is cloned via NheI/XhoI), so it cannot be screened by Digest A and does not share a lane with the four plasmids. Once the engineered PENK is cloned into an expression backbone, that construct will need its **own restriction map** computed fresh.

2b. Engineered PENK therapy plasmid — in silico build + QC

The four native plasmids are the **test-bed**; the **engineered PENK is the intended therapeutic construct**. To make it a plasmid rather than a linear fragment, the engineered insert (NheI–Kozak–CDS–XhoI, 885 bp) was cloned in silico into the **pcDNA3.1(+)** backbone shared by the test-bed vectors.

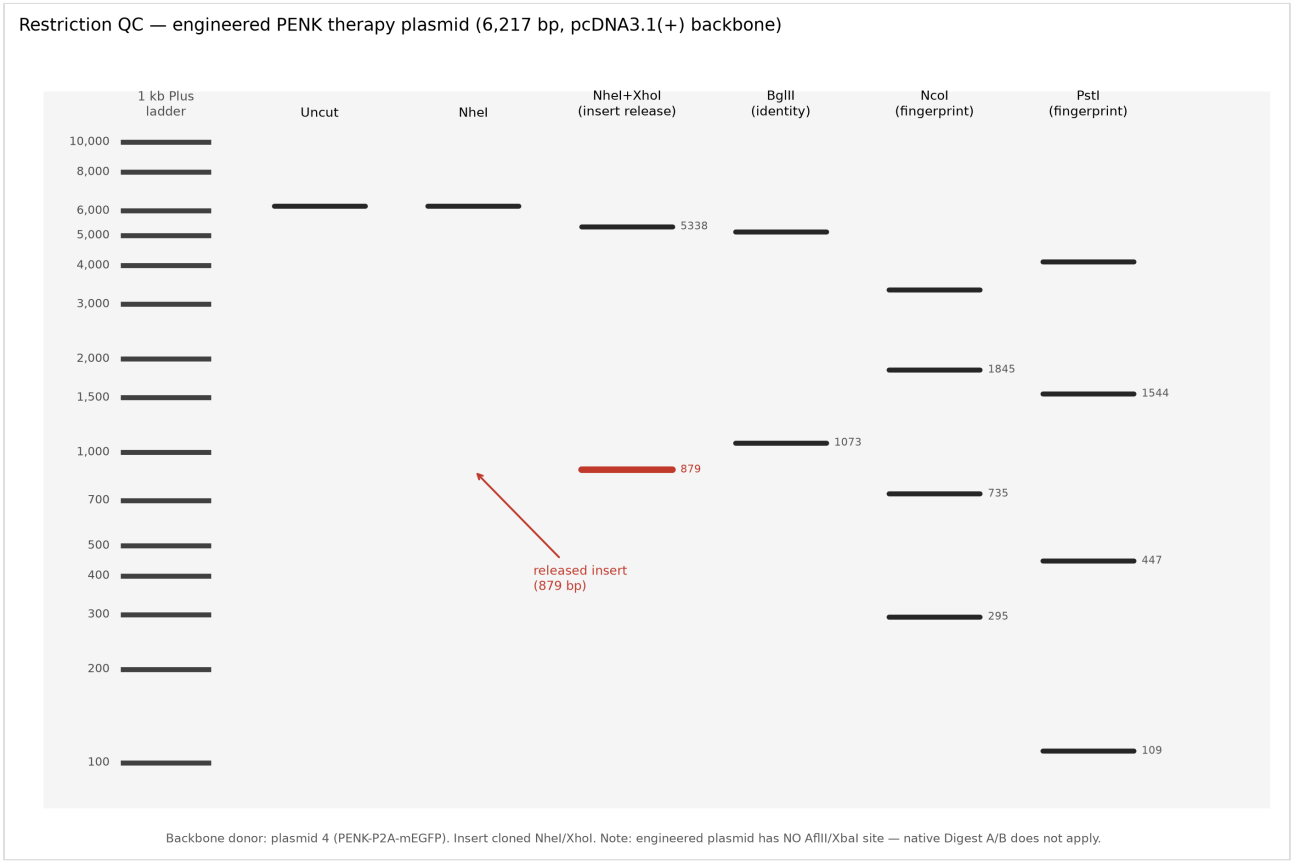
Assembly. Backbone donor = plasmid 4 (PENK-P2A-mEGFP), chosen for its **single** NheI and **single** XhoI sites (a clean directional swap). The native P2A–mEGFP cassette (1,639 bp, NheI→XhoI) was excised and replaced by the engineered insert, giving a **6,217 bp circular plasmid**. Verified: single NheI + single XhoI, engineered CDS present exactly once (288 aa, **all 7 YGGFEVVG cores**), native cassette removed. Promoter (CMV), polyA (bGH), and selection (NeoR/AmpR) are held identical to the test-bed, so the therapy construct stays parallel to the validated expression system.

Purpose-built QC. The native Digest A/B scheme does not transfer — the engineered plasmid has **no AflII site**, and its only XbaI site is in the backbone. Five diagnostic digests replace it:

Digest	Enzyme(s)	Expected bands (bp)	Purpose
A — insert release	NheI + XhoI	**5,338 + 879**	insert present & correct size
B — linearize	NheI	6,217	total size
C — identity	BglII	5,144 + 1,073	empty vector loses the 1,073 band
D — fingerprint	NcoI	3,342 + 1,845 + 735 + 295	rearrangement check

Digest	Enzyme(s)	Expected bands (bp)	Purpose
E — fingerprint	PstI	4,117 + 1,544 + 447 + 109	independent fingerprint

The virtual gel (`engPENK_therapy_qc_gel.png`) shows all five digests vs a 1 kb-plus ladder; the protocol and bench recipe are in `engPENK_therapy_qc_protocol.md`, and the annotated plasmid is `engPENK_therapy_plasmid.gb`.



Virtual restriction gel — engineered PENK therapy plasmid (6,217 bp). Five diagnostic digests; the 879 bp NheI+XhoI insert-release band (red) is the key signature.

Status. In silico design. After bench cloning, Sanger sequencing across the CDS is mandatory — the seven cores are synonymously distinct by design and a gel cannot verify them (the engineered analogue of the gel-invisible plasmid-1 truncation).

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3. Human anti-NEP (MME) siRNA design

Rationale. Neprilysin (NEP / CD10, gene **MME**) is the principal enkephalin-degrading peptidase. Silencing it prolongs the half-life of the enkephalins produced by the PENK construct — the knockdown arm of the therapy.

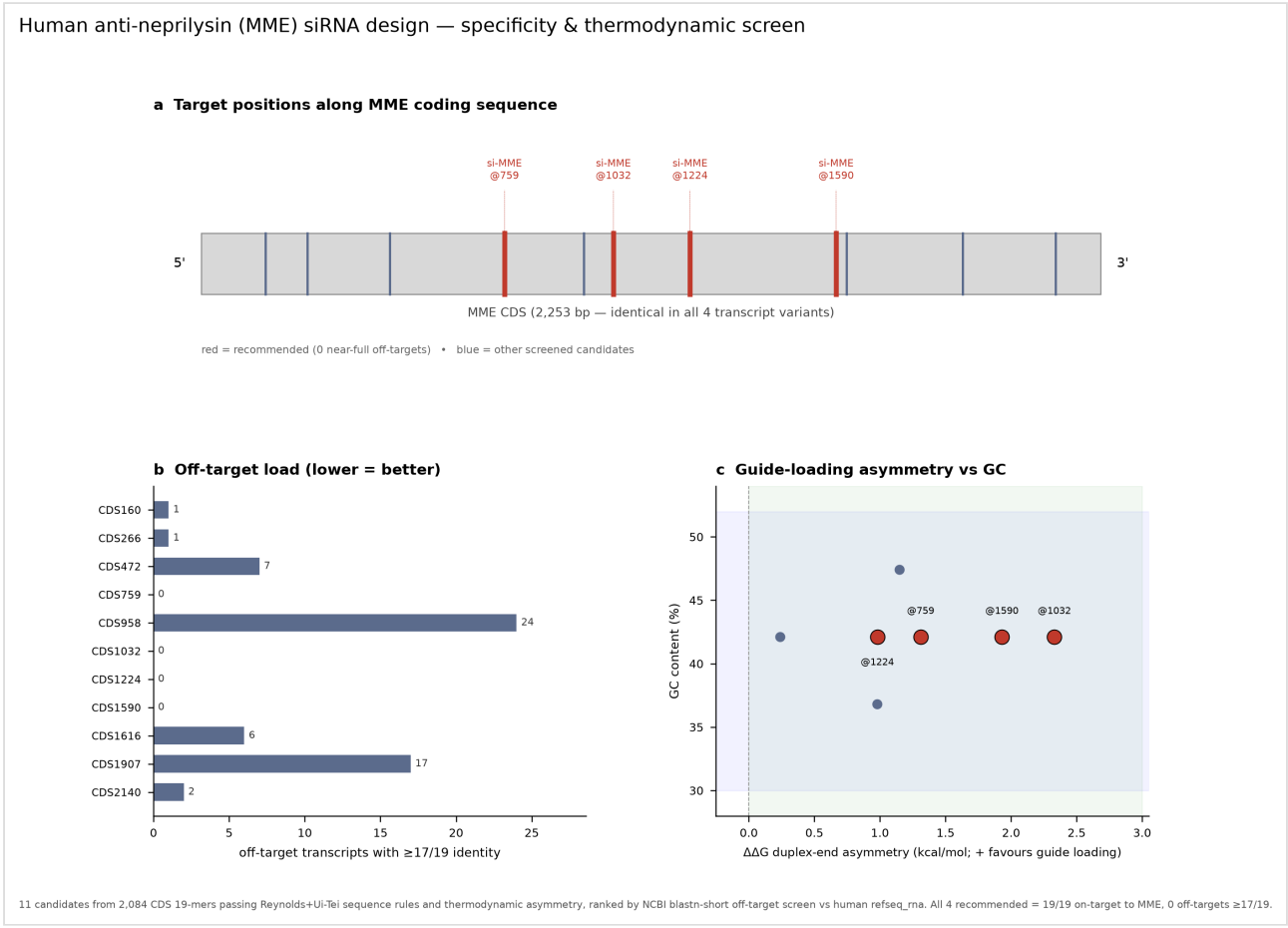
Target. All four human MME transcript variants (NM_000902.5, NM_007287/288/289) share an **identical 2,253 bp CDS**; every design targets that shared CDS, so a single siRNA silences all isoforms.

Pipeline. 2,084 candidate 19-mers enumerated across the CDS → **228** pass sequence rules (Reynolds + Ui-Tei: GC 30–52%, positional base preferences, no long homopolymer/GC runs) → ranked by thermodynamic duplex-end asymmetry (favouring guide-strand RISC loading) → off-target screened by NCBI blastn-short against human `refseq_rna`.

Four recommended duplexes (all 19/19 on-target, 0 off-targets ≥17/19; standard 3′ dTdT overhangs):

#	CDS pos	Guide / antisense (5'→3')	GC%	ΔΔG	worst off-target
si-MME-1	759	UUCCUGACGAAUCAAUCUGdTdT	42.1	+1.31	16/19
si-MME-2	1032	UUCUGGAGCAUAAACAACCdTdT	42.1	+2.33	16/19
si-MME-3	1224	AUAGUUUGCACAACGUCUCdTdT	42.1	+0.98	14/19
si-MME-4	1590	UAUCCACUCAUCUUUGUCCdTdT	42.1	+1.93	16/19

Passenger strands, mRNA positions, and full scoring are in `sirna_design_table.csv`; the design figure is `sirna_design_figure.png`.



Anti-MME siRNA panel — target positions on the MME CDS, off-target load, and guide-loading asymmetry vs GC.

Validation recommended: qPCR of MME at 48 h (expect ≥70% knockdown for ≥2 duplexes) + NEP activity/CD10 Western; use ≥2 independent effective siRNAs to control for off-target phenotypes. A context-dependence caveat from the project bibliography is recorded — this design provides tool reagents, not a settled biological strategy.

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4. Lipoplex coarse-grained MD

Question. Does the ionizable lipid's protonation state control DNA binding as the delivery design assumes — weak binding at physiological pH (neutral, for circulation), strong binding at endosomal pH (protonated, for cargo condensation)?

Design. MARTINI 2 CG-MD of a 60-bp CG B-DNA duplex with three lipid formulations, each cationic-lipid 120 : DOPE 80 : cholesterol 40 in ~24k CG waters, 150 mM NaCl:

- **DOTAP** — permanent +1 (control for constitutive binding)
- **MC3-H⁺** — ionizable MC3 surrogate, protonated (endosomal-pH state)
- **MC3⁰** — same lipid, neutral (physiological-pH state)

3 independent velocity-seed replicates per formulation (9 runs), analyzed with mean ± SEM.

Final results (200 ns × 3 seeds)

All 9 replicates were extended from 50 ns to **200 ns** (each continued from its checkpoint); metrics use the equilibrated second half (100–200 ns).

Formulation	DNA–lipid contacts	Min DNA–lipid dist (nm)	RDF peak g(r)	DNA Rg (nm)
DOTAP (permanent +)	24.7 ± 5.5	0.460 ± 0.003	1.54 ± 0.12	4.10 ± 0.13
MC3-H⁺ (endosomal pH)	34.4 ± 8.6	0.469 ± 0.013	1.57 ± 0.31	4.24 ± 0.07
MC3⁰ (physiological pH)	8.1 ± 4.9	1.270 ± 0.563	1.04 ± 0.09	4.18 ± 0.002

The pH switch is reproduced, and the extension confirms it robustly. At endosomal pH the protonated ionizable lipid (MC3-H⁺) binds DNA as tightly as the permanent-cation control DOTAP — both reach ~0.47 nm closest approach — condensing and protecting the cargo. At physiological pH the neutral form (MC3⁰) holds DNA ~1.3 nm away, with an RDF peak at the bulk value (~1.0) and one seed that never binds (0.08 contacts, 2.36 nm) — the release/circulation state.

Convergence — the extension did its job

At 50 ns the between-seed spread was large **because binding is a stochastic nucleation event that had not finished** (cumulative-mean contacts still rising, first-half → second-half drift up to +200%, while within-replicate block error stayed small at 0.7–4.2 contacts — i.e. between-seed sampling, not thermal noise). The 200 ns extension resolved this:

- **MC3-H⁺ error bars collapsed** — contacts SEM 15.1 → 8.6, min-distance SEM 0.126 → 0.013 nm (≈10× tighter). The short run had caught nucleation mid-transient; the long run confirms genuine strong binding.
- **DOTAP** stayed strong and tightened further (contacts SEM 6.1 → 5.5) — constitutive binding, as expected for a non-titratable lipid.
- **MC3⁰** stayed weak throughout — the intended physiological-pH control.

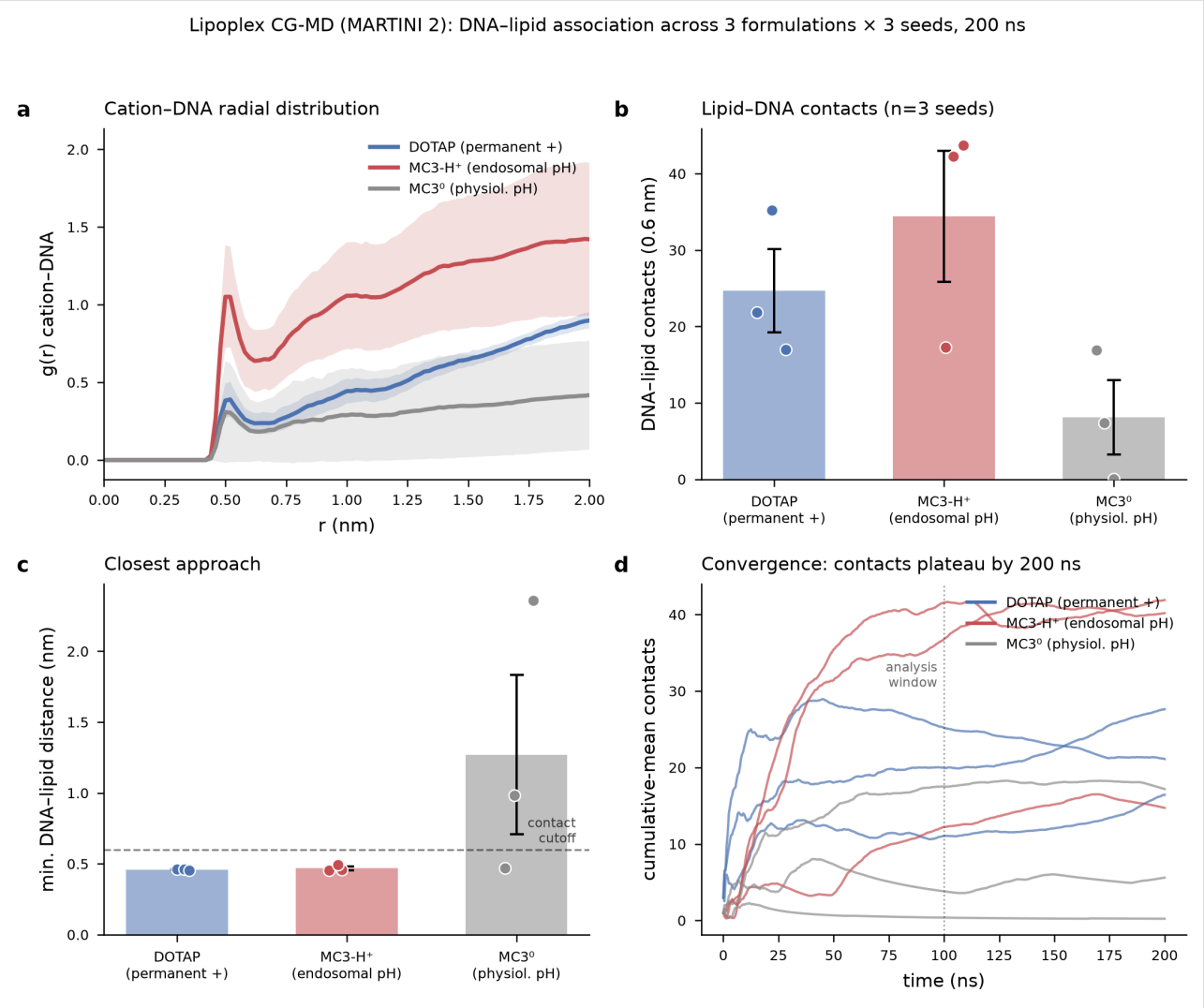
Cumulative-mean contacts plateau by ~100 ns for all formulations, so the 100–200 ns analysis window is equilibrated. This was a genuine convergence requirement, not padding.

Honest limitation (stated by design)

The DOTAP, MC3-H⁺ and MC3⁰ topologies are **MARTINI-2 analogy models** (built from DOPC/DUPC), **not** independently validated published parameters. Consequently all conclusions are **qualitative / comparative** — the relative ranking of DNA-condensing tendency across formulations and protonation states — **not** quantitative binding free energies. More sampling improves convergence but cannot fix a parameter-provenance limitation; this is disclosed rather than hidden.

Figure: `lipoplex_md_200ns.png` (RDF with replicate spread · contacts with individual seeds · closest approach · convergence). Per-replicate and summary metrics in

`md_formulation_summary_200ns.json`; full 50→200 ns convergence tables in `04_lipoplex_md/RESULTS_200ns.md`. The Stage-1 50 ns figure/metrics are retained (`lipoplex_md_50ns.png`, `md_formulation_summary.json`) for provenance.



Lipoplex CG-MD (200 ns × 3 seeds): cation–DNA RDF, DNA–lipid contacts, minimum distance, and cumulative-mean convergence. Mean ± SEM across replicates.

4b. Therapeutic delivery rationale — putting the two arms together

The four workstreams are not independent analyses; they are components of one system whose therapeutic logic is **endogenous opioid enhancement**: make more enkephalin, degrade less of it, at the pain relay.

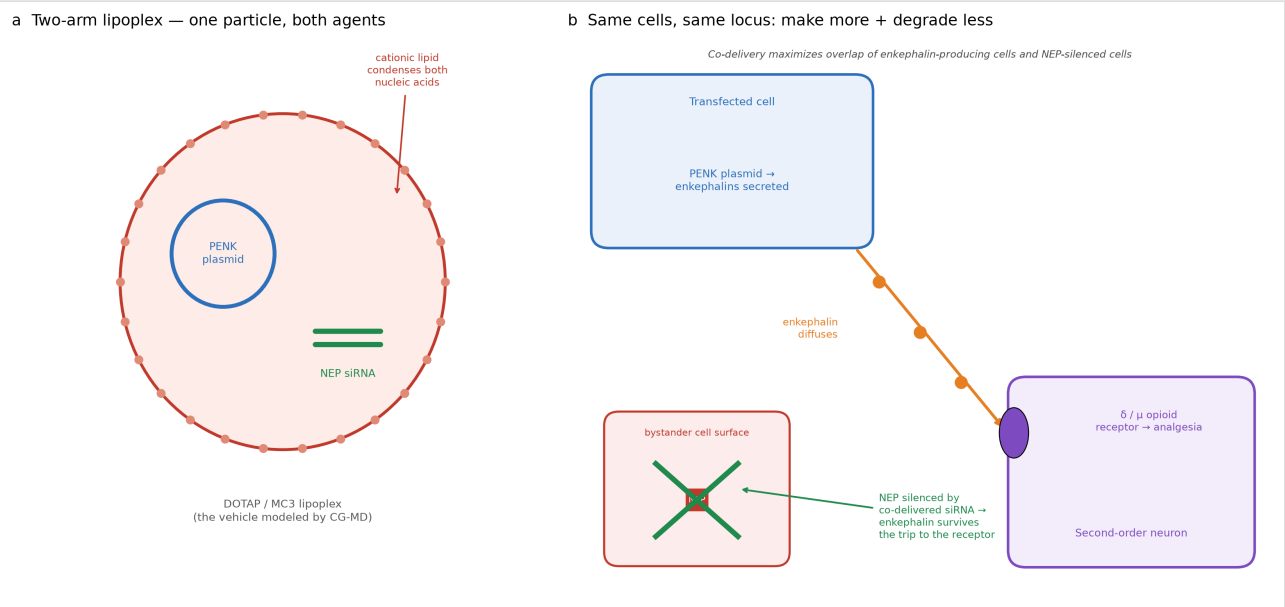
Mechanism. The PENK construct produces proenkephalin, which is processed to enkephalins — endogenous opioid peptides acting on δ- (and μ-) opioid receptors to produce opioid-like analgesia without an exogenous opioid. Neprilysin (NEP / MME / CD10) is the principal enkephalin-degrading ectopeptidase, inactivating enkephalins in the synaptic space within seconds. Silencing NEP prolongs enkephalin action — the knockdown arm. The two arms are synergistic.

Where the siRNA goes — co-formulation, not just co-expression. NEP acts as a cell-surface ectoenzyme and enkephalin is degraded extracellularly, but siRNA silencing is **cell-autonomous** — it only affects the cells actually transfected. Secreted enkephalin then diffuses

past the surfaces of many neighboring cells, and NEP on any of them will degrade it. Silencing NEP only in the enkephalin-producing cell is therefore insufficient. The design consequence: **co-formulate the NEP siRNA and the PENK plasmid in the same lipoplex, delivered to the same locus**, so the population of enkephalin-producing cells overlaps maximally with the population of NEP-silenced cells. This is precisely why the lipoplex CG-MD (§4) is load-bearing — the lipoplex is the vehicle that carries both nucleic acids into the same region, and the siRNA off-target screen (§3) is part of that same package.

Where in the CNS. Lipoplexes do not cross the blood–brain barrier, so delivery is local/regional to the analgesic circuit — principally the **spinal dorsal horn (intrathecal)**, the first nociceptive synapse and rich in δ/μ opioid receptors; supraspinal targets (PAG, RVM) are alternatives. There is clinical precedent for the enkephalin-gene-therapy-for-pain concept (an HSV proenkephalin vector reached trials for cancer pain); the additions here are the NEP-knockdown arm and a non-viral lipoplex vehicle.

The two-arm concept and its dorsal-horn mechanism are shown in `therapy_delivery_schematic.png`.



Therapeutic delivery rationale. (a) NEP siRNA and PENK plasmid co-formulated in one DOTAP/MC3 lipoplex. (b) Co-delivery to the same locus maximizes overlap of enkephalin-producing and NEP-silenced cells; delivered to the spinal dorsal horn.

Honest caveats. (i) The cell-autonomy mismatch is the central design risk — co-formulation plus local delivery mitigates but does not erase it. (ii) NEP has other substrates (substance P, bradykinin, natriuretic peptides, amyloid-β); broad chronic knockdown could impair Aβ clearance, so the knockdown is kept **local and transient** by using siRNA/lipoplex rather than a permanent vector. (iii) Sustained opioid tone raises tolerance/dependence considerations. (iv) siRNA and non-integrating plasmid expression are transient — adequate for proof-of-concept, but a chronic indication would want a durable vector.

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5. What each deliverable establishes

Workstream	Establishes	Actionable output
Plasmid audit	Which construct to use (plasmid 3); which is defective (plasmid 1)	Use plasmid 3; Sanger-verify or discard plasmid 1
Restriction QC	Identity/gene-order check before transfection	2-digest protocol + virtual gel
siRNA design	Knockdown reagents for the NEP arm	4 orderable duplexes + validation plan

Workstream	Establishes	Actionable output
Lipoplex CG-MD	pH-dependent DNA binding of the delivery vehicle	Formulation ranking + convergence-backed screen

6. Reproducibility

Every tool version, force field, database accession, and published design rule is recorded in `methods_references.md`. The full analysis code, input topologies, protocols, and figures are versioned in the accompanying Git repository. The MD parameter-provenance caveat is stated in both the methods log and the results above.

This brief reports the final 200 ns MD results (all 9 replicates converged). The optional Boltz peptide–receptor co-fold (μ/δ opioid receptor × native/engineered enkephalin) was attempted opportunistically but did not complete — it hit a cluster home-directory disk-quota limit during MSA generation, so no co-fold structures were produced. It is not one of the core deliverables and can be re-run once quota is freed.